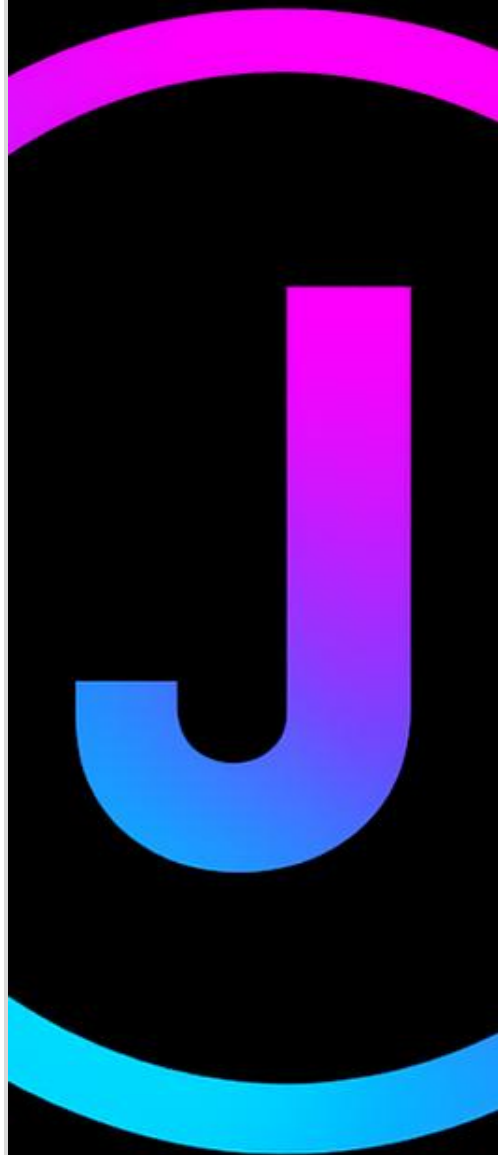




HUB

JoyDeck Hub user manual

By DGVeLab

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1. Introduction

JoyDeck is an advanced control system composed of a dedicated hardware device and a desktop application (JoyDeck Hub), designed to transform an 8-direction physical joystick into a powerful tool for automation, control, and interaction with the operating system.

Unlike a standard controller or macro pad, JoyDeck is built with a technical and modular approach: each joystick direction can be assigned to a distinct, fully configurable action, with immediate visual feedback through RGB LEDs and on-screen icons.

The project is intended for **advanced users**, makers, professionals, and power users who want to:

- speed up repetitive operations
- control applications, media, and the system without using the keyboard
- create complex and contextual macros
- integrate custom hardware into their workflow

JoyDeck Hub communicates with the device via USB serial connection and delegates to the ESP32 the management of the joystick, LEDs, and hardware states, while keeping all action-execution logic on the PC.

This manual aims to provide **technical yet practical documentation**, guiding the user from the first installation to advanced use of individual Actions, with real examples and concrete use cases.

2. Terms of Use and Safety

JoyDeck Hub is software that interacts directly with the operating system, simulating keyboard and mouse input, launching applications, opening web links, and executing system commands. For this reason, it must be used responsibly.

General Warnings

- Improper use of Actions may cause applications to close, interrupt processes, or lead to loss of unsaved data.
- The Simulate Key, System Control, and Media Control features operate at a global system level: make sure you fully understand their behavior before using them permanently.
- JoyDeck does not implement sandboxing mechanisms: actions are executed with the same permissions as the user running the program.

System Security

- Some antivirus tools or Windows protection systems (SmartScreen, Defender) may flag JoyDeck Hub as an unsigned application. This is normal for software distributed without a digital certificate.
- Always verify the source of the program and use only official versions.
- Avoid importing configuration files from untrusted sources.

User Responsibility

The user is solely responsible for the configurations created and the actions assigned to the device. The software author is not liable for any damage resulting from improper use or from use that does not comply with the guidelines provided in this manual.

It is recommended to:

- test each Action using the Test button before real use
- avoid assigning critical commands without a confirmation mechanism
- always maintain control of the device during operation

Using JoyDeck implies acceptance of these terms. These warnings cover the use of system commands, key simulation, automations, and user responsibility.

3. JoyDeck Device Overview

JoyDeck is a *standalone physical* input device designed for one-handed use and to provide immediate, tactile, and visual interaction with the system. The philosophy behind the project is to *reduce the number of steps between intention* and action by delegating command selection and activation to the physical joystick.

3.1 Main components

The device is composed of the following functional elements:

- **2-axis analog joystick**
Used for directional input detection. The firmware interprets movement across 8 logical directions:
 - UP, DOWN, LEFT, RIGHT
 - UP-LEFT, UP-RIGHT, DOWN-LEFT, DOWN-RIGHT
- **Joystick-integrated push button**
The central button allows:
 - command confirmation (On-confirm mode)
 - single, double, or long press
 - exiting special modes (Mouse mode, Simulate Key mode, Media mode)
- **Addressable RGB LED ring**
Provides immediate visual feedback on system status:
 - selected direction
 - command execution
 - active mode
 - error or invalid command
- **ESP32 microcontroller**
Handles:
 - joystick reading
 - LED control
 - serial communication with the JoyDeck Hub
 - basic logic and hardware states
- **USB connection**
Used both to power the device and to enable data communication with the computer.

3.2 Separation of Roles: Hardware vs Software

A fundamental aspect of JoyDeck is the clear separation of responsibilities:

- **ESP32 (hardware)**
 - detects movements and button presses;
 - manages LEDs in real time;
 - sends raw events (axes, button, state) to the PC;
 - remains simple, stable, and responsive.
- **JoyDeck Hub (software)**
 - interprets movements;
 - applies filters (dead zone, hysteresis, diagonals);
 - decides when and how to execute a command;
 - simulates system input;
 - manages advanced modes and automation.

This architecture makes JoyDeck:

- flexible;
 - upgradable;
 - easily extensible.
-

3.3 Direction Logic

The 8 directions are not simple static zones; they are dynamically determined through:

- central dead zone
- stability thresholds
- intelligent diagonal handling
- hysteresis filters

This approach prevents:

- unintentional activations
- bouncing between directions
- unstable selections during fast movements

Each direction is mapped one-to-one with a row in the Commands tab inside JoyDeck Hub.

3.4 Visual Feedback as a Language

The LED feedback is not merely aesthetic; it forms a true visual language designed to provide immediate information about the system state without requiring attention to the screen.

The LED behavior follows a coherent and easily recognizable logic:

- **Illuminated LED segment** → currently active direction
- **All LEDs on** → command executed or persistent mode
- **LEDs off** → no command assigned or neutral state
- **Red color** → error or command not executed correctly

This approach allows JoyDeck to be used without looking at the screen, keeping attention on the primary activity and making the device effective even in dynamic or professional contexts.

3.5 Special Operating Modes

The device can enter dedicated modes, identifiable both by their behavior and by the LED feedback:

- **Mouse Mode** – the joystick controls the cursor
- **Media Controller Mode** – the joystick manages media playback
- **Spotify Controller Mode** – advanced contextual control
- **Simulate Key Mode** – handling of persistent or rapid-fire key presses

These modes are activated by the software, but they are always mediated by the physical device, which remains the central control point.

3.6 Usage Philosophy

JoyDeck does not replace keyboard and mouse, but complements them.

It is designed for:

- *frequent actions;*
- *complex shortcuts;*
- *operations that benefit from a physical gesture;*
- *reduction of cognitive load.*

The device becomes effective when configured according to the user's real habits, not as a universal tool but as a personalized extension of the workflow.

4. Installation and First Launch

This chapter describes in detail the operations required to make JoyDeck operational for the first time, from system preparation to confirming the correct functioning of the device.

The installation is designed to be simple and fast, but it is important to follow the steps in the indicated order to avoid communication or initial configuration issues.

4.1 System Requirements

To use JoyDeck Hub you need:

- Windows 10 or Windows 11 operating system
- Available USB port
- Standard user permissions (administrator privileges are not required)
- JoyDeck device based on ESP32

USB–Serial Driver

For proper device recognition, installation of the CP210x USB–serial driver is required.

- Required driver: CP210x Universal Windows Driver
- The driver is available in the official JoyDeck GitHub repository, along with the device firmware.

Without this driver:

- the device may not appear as a COM port;
- the serial connection with JoyDeck Hub will not be possible.

Note

Device firmware installation and updates will be covered in a dedicated chapter, separate from the first-launch procedure.

4.2 Connecting the Device

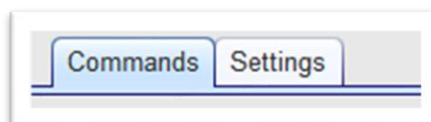
1. Connect JoyDeck to the computer using a USB cable.
2. Wait for Windows to automatically recognize the device.
3. Verify that the device appears as a serial (COM) port in the system.

During this phase:

- LEDs may briefly turn on;
- the joystick is not yet operational;
- no commands are executed.

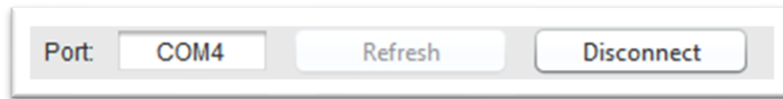
4.3 Launching JoyDeck Hub

1. Start **JoyDeck Hub** using the executable or the shortcut created during installation.
2. On first launch, the program loads the basic configuration.
3. The main interface is displayed with the **Commands** and **Settings** tabs.



If JoyDeck Hub is configured for automatic startup, the program may run in the background without immediately showing the window.

4.4 Serial Port Detection and Connection



JoyDeck Hub uses a serial connection to communicate with the device.

- The software can **automatically detect** the serial port associated with JoyDeck by clicking the **Refresh button**.

Alternatively, the port can be selected manually by typing the port name directly into the dedicated textbox.

Once the port is selected:

1. Press **Connect**.
2. Wait for the connection confirmation.
3. The button changes state and the settings become editable.

In case of error:

- verify that the device is connected;
- check that the port is not being used by other programs;
- repeat the procedure.

4.5 Initial Calibration

At the first connection, JoyDeck automatically performs a joystick-center calibration.

During calibration:

- do not touch the joystick;
- keep the device still;
- wait for the automatic completion of the procedure.

Calibration serves to:

- determine the joystick resting position;
- compensate mechanical tolerances;
- ensure precise direction selection.

At the end of calibration:

- JoyDeck enters the **ready** state;
- LED feedback indicates system availability;
- the device is operational.

4.6 Initial System State

At first launch, JoyDeck uses a basic configuration:

- no critical Actions assigned;
- default trigger mode;
- standard sensitivity parameters.

This state is designed to allow:

- a learning phase;
- gradual device configuration;
- safe testing without risks for the system.

It is recommended to customize Actions only after understanding the joystick's basic behavior.

4.7 Common Issues at First Launch

- No port detected: disconnect and reconnect the device, then restart JoyDeck Hub.
- Connection failed: verify that no other software is using the same serial port.
- Inaccurate movements: repeat calibration while keeping the joystick still.

The FAQ section covers these and other cases in more detail.

5. Device Usage (Hardware)

This chapter describes the daily use of the JoyDeck device from a physical perspective, focusing on direct interaction with the joystick, button, and LED feedback, regardless of the Actions configured in the software. The goal is to provide a clear understanding of how JoyDeck responds to user movements and inputs, enabling conscious and predictable device operation.



5.1 Rest State and Neutral Position

When JoyDeck is connected and the system is ready:

- the joystick rests in the **central (neutral) position**;
- no direction is active;
- LEDs remain off, unless specific modes or configurations are enabled.

The neutral position represents the device's safety state: no Action is selected or executed.

5.2 Joystick Movement and Direction Selection

When the joystick is moved away from the central position, JoyDeck interprets the movement using an **8-direction** logic.

During movement:

- the direction is selected only when the joystick exceeds the dead zone;
- light or involuntary movements are ignored;
- the selection becomes stable only beyond a certain threshold.

This behavior prevents accidental activations and ensures precise direction selection even during fast movements.

5.3 Cardinal and Diagonal Directions

JoyDeck automatically distinguishes between:

- **Cardinal directions:** UP, DOWN, LEFT, RIGHT
- **Diagonal directions:** UP-LEFT, UP-RIGHT, DOWN-LEFT, DOWN-RIGHT

Diagonal directions are recognized only when:

- movements along the X and Y axes are sufficiently similar;
- the joystick is held steadily in position.

This prevents unintentional switching between cardinal and diagonal directions during use.

5.4 LED Feedback During Use

LED feedback accompanies the user during every phase of interaction:

- **Illuminated LED segment** → selected direction
- **All LEDs on** → command executed or active mode
- **LEDs off** → joystick in neutral position

The visual feedback allows the user to understand the device state even without looking at the screen.

5.5 Releasing the Joystick

When the joystick is released and returns toward the central position:

- the selected direction is considered completed;
- depending on the trigger mode, the command may be executed;
- LED feedback is updated accordingly.

Returning to center is a fundamental event, used by the system to:

- confirm or cancel an action;
 - exit temporary states;
 - ensure consistent behavior between one command and the next.
-

5.6 Joystick Button

The button integrated into the joystick is an active part of the hardware interaction and can generate different types of input:

- **Single click**
- **Double click**
- **Long press**

The meaning of these inputs depends on the active mode and software configuration.

The button is mainly used to:

- confirm a selected command;
 - exit special modes;
 - provide an additional input without moving the joystick.
-

5.7 Safety Behaviors

JoyDeck integrates several hardware-level safety behaviors:

- no Action is executed if the joystick does not exceed the minimum threshold;
- direction selection is blocked during a click;
- the system requires a return to center before accepting new inputs.

These measures reduce the risk of accidental executions and improve device predictability.

5.8 Ergonomic Use

For optimal use, it is recommended to:

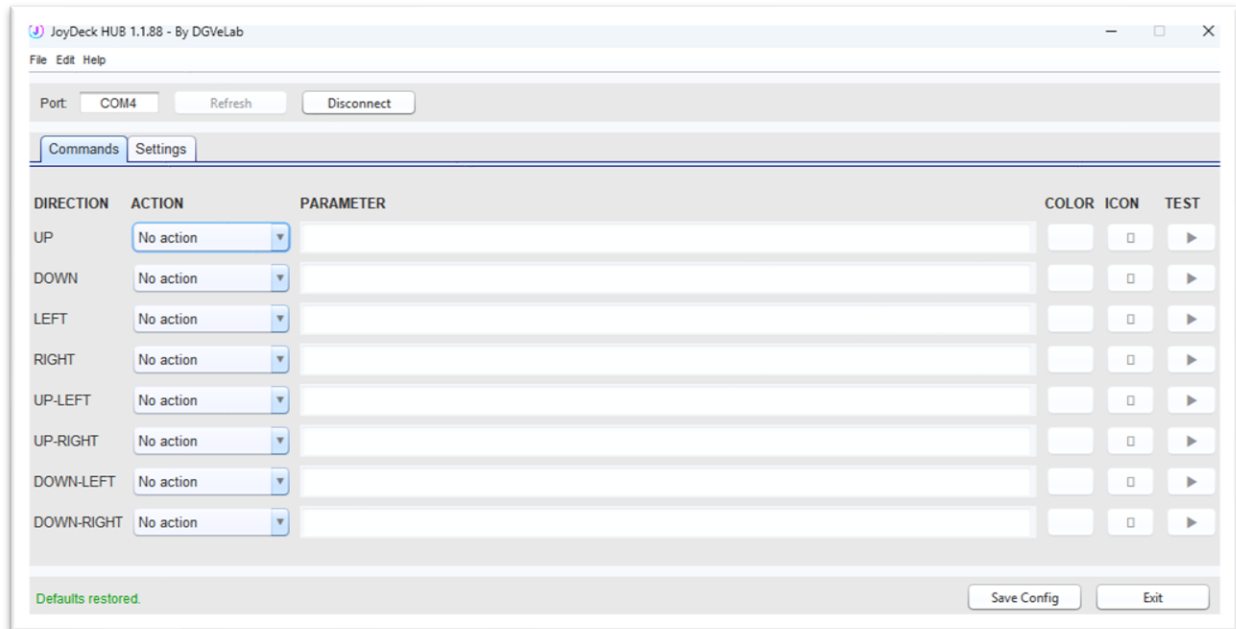
- keep the device stable on a flat surface;
- use firm but controlled movements;
- avoid excessive lateral pressure on the joystick;
- become familiar with the directions before assigning critical Actions.

JoyDeck's effectiveness increases significantly when the physical gesture becomes natural and repeatable.

6. Graphical Interface – Commands

The Commands tab represents the operational core of JoyDeck Hub. From this interface, each joystick direction can be associated with a specific Action, defining its behavior, parameters, and visual feedback.

Configuration is always performed per direction: each row of the tab corresponds to one of the 8 logical joystick directions.



6.1 General Structure of the Commands Tab

The Commands tab is organized into rows and columns. Each row represents a direction, while the columns allow configuration of the associated command.

The main elements are:

- Direction – indicates the joystick direction (UP, DOWN, LEFT, RIGHT, diagonals)
- Action – type of action to execute
- Value – parameter associated with the Action
- Color – LED feedback color
- Icon – icon displayed on screen during selection
- Test – button to test the command

6.2 Directions and Mapping

The available directions are always eight and follow a fixed mapping:

- UP
- DOWN
- LEFT
- RIGHT
- UP-LEFT
- UP-RIGHT
- DOWN-LEFT
- DOWN-RIGHT

This mapping is consistent across:

- the physical device;
- LED feedback;
- the Commands tab;
- the internal software logic.

It is not possible to modify the number or position of the directions: customization occurs exclusively through Actions.

6.3 Action Column

The **Action** column allows selecting the type of command to associate with the direction.

Each Action defines:

- what is executed;
- how JoyDeck interprets the movement;
- whether the command is instantaneous or persistent;
- whether it activates a special mode.

Available Actions include, among others:

- No Action
- Open App
- Open Link
- System Control
- Simulate Key
- Mouse Move
- Orbital
- Media Control
- Spotify
- HTTP Request

Each Action is described in detail in a dedicated chapter.

6.4 Value Column

The **Value** column contains the parameter required by the selected Action.

The meaning of the field varies depending on the Action type:

- file or application path;
- URL or link;
- key sequence;
- system command;
- specific configuration (media, mouse, network).

The Value field may be:

mandatory;

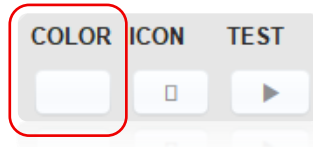
optional;

ignored, depending on the chosen Action.

An invalid or empty Value may cause:

- failure to execute the command;
- an error notification;
- negative LED feedback.

6.5 Color Column



The **Color** column allows associating a color with the command.

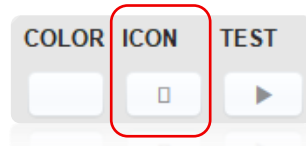
The color is used to:

- visually identify the command;
- provide feedback during selection;
- indicate execution status or active mode.

It is advisable to:

- use consistent colors for similar commands;
- avoid colors that are too similar for adjacent directions;
- reserve red for error indications.

6.6 Icon Column



The **Icon** column allows associating a graphical icon with the command.

During use:

- the icon is displayed on screen when the direction is selected;
- it provides immediate feedback on the associated command;
- it reduces the need to look at the Commands tab.

Icons are automatically normalized by the software to ensure visual consistency.

6.7 Test Button



The **Test** button allows executing the associated Action without using the physical device.

It is useful for:

- verifying the correctness of the Value;
- safely testing critical commands;
- checking the Action's behavior before real use.

Testing does not modify the joystick state and does not require hardware movement.

6.8 Unassigned Commands

When a direction is set to No Action:

- no command is executed;
- LEDs remain off;
- the direction can still be selected.

This configuration is useful for:

- reserving directions for future use;
- preventing accidental executions;
- simplifying the learning phase.

6.9 Saving Changes

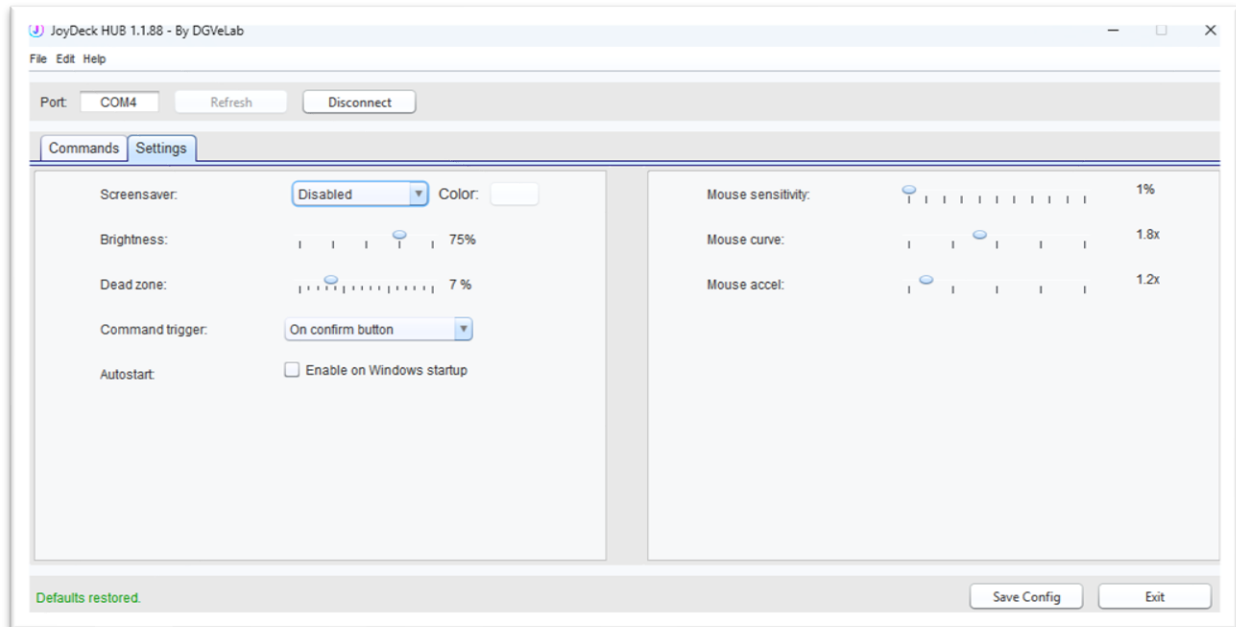
Changes made in the Commands tab become effective only after saving.

It is recommended to:

- save frequently;
- test each Action individually;
- proceed incrementally during configuration.

7. Graphical Interface – Settings

The Settings tab allows configuring the global behavior of JoyDeck. Unlike the Commands tab, which acts on individual directions, the options in this section affect the entire system, both hardware and software. Proper configuration of the Settings is essential to adapt JoyDeck to your usage style and ensure consistent and predictable device behavior.



7.1 General Structure of the Settings Tab

The Settings tab gathers all options that do not depend on a single direction, but instead regulate:

- joystick sensitivity;
- trigger behavior;
- global visual feedback;
- special modes;
- integration with the operating system.

Changes to the Settings take effect immediately after saving and are transmitted to the device.

7.2 LED Brightness and Feedback

This section allows adjusting the global behavior of the LEDs:

- Brightness – sets the overall brightness of the LED ring;
- Screen / Idle color – color used in idle mode or screensaver mode.

Brightness that is too high may be distracting in low-light environments, while values that are too low may reduce feedback visibility.

7.3 Joystick Dead Zone

The Dead zone parameter defines the central area of the joystick within which movements are ignored.

This setting helps to:

- compensate for mechanical tolerances of the joystick;
- avoid unintentional activations;
- improve stability at rest.

A dead zone value that is too low may cause unwanted selections, while a value that is too high reduces device responsiveness.

7.4 Command Trigger Modes

JoyDeck supports several command execution modes, selectable from the Settings:

- On release – the command is executed when the joystick is released;
- On confirm – the command is selected with the joystick and confirmed via button;
- Hold – the command is executed only if the direction is held for a defined time.

These modes significantly change the user experience and should be chosen based on the type of assigned Actions.

7.5 Mouse Mode

When a direction is configured with the Mouse Move Action, three global parameters become active to regulate cursor behavior:

- Mouse Sensitivity

Determines cursor movement speed. Higher values produce faster movements for the same joystick displacement.

- Mouse Curve

Modifies cursor response based on movement intensity. A smoother curve makes small movements more precise, while higher values favor speed during larger movements.

- Mouse Accel

Introduces progressive acceleration: by holding the joystick in a direction, cursor speed increases over time.

These parameters take effect only when Mouse mode is active and do not influence other Actions.

7.6 Automatic Startup and System Integration

JoyDeck can be configured to:

- start automatically when the system boots;
- run in the background;
- reduce visual impact during daily use.

These options improve workflow integration, especially in professional environments.

7.7 Saving and Restoring Settings

Changes to the Settings must be saved to become permanent.

It is recommended to:

- modify one parameter at a time;
- test joystick behavior;
- save only after verifying system stability.

In case of abnormal behavior, it is always possible to restore a previous configuration.

8. Actions

This chapter represents the functional core of JoyDeck.

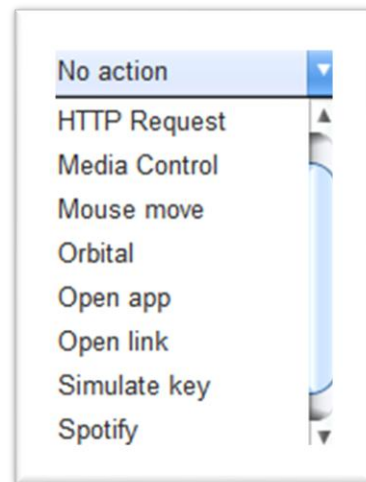
Actions define what happens when a joystick direction is activated. Each Action embodies a specific behavior and can range from a simple operation (opening an application) to complex automation and system-control scenarios.

DIRECTION	ACTION	PARAMETER	COLOR	ICON	TEST
UP	No action				

List of Available Actions

The Actions currently available in JoyDeck Hub are:

1. No Action
2. Open App
3. Open Link
4. System Control
5. Simulate Key
6. Mouse Move
7. Orbital
8. Media Control
9. Spotify
10. HTTP Request



The following chapters analyze each Action in depth, maintaining a level of detail suitable both for beginners and for users who want to leverage JoyDeck in advanced ways.

8.1 No Action

Description

No Action indicates that no operation is associated with a joystick direction. The direction remains valid from a physical standpoint, but it produces no effect on the system.

This Action represents an intentionally neutral and safe state.

When to use it

No Action is recommended when:

- you want to leave one or more directions unused;
- you are configuring JoyDeck gradually;
- you want to avoid any accidental execution;
- a direction is reserved for future use.

It is the default configuration at first launch.

Parameters (Value)

The **No Action** does not use the Value field.

Any content in the field is ignored.

Joystick behavior

When a direction configured as No Action is selected:

- the joystick correctly recognizes the direction;
- no command is executed;
- releasing the joystick has no effect on the system.

The behavior is identical regardless of the trigger mode.

Visual feedback

With No Action:

- LEDs remain off;
- no icon is displayed;
- no execution feedback is shown.

This makes inactive directions immediately recognizable.

Practical examples

Basic example

Leave diagonal directions set to No Action to reduce initial complexity.

Organizational example

Use No Action to logically separate groups of commands (e.g., only cardinal directions active).

Common mistakes

- Confusing No Action with a device malfunction.
 - Attempting to use the Value field with this Action.
-

Notes and tips

No Action is a useful tool for designing a clean and progressive configuration.

It is recommended to always start from No Action and add necessary Actions one at a time.

8.2 Open App

Description

The Open App Action allows launching applications, executables, scripts, or system URIs directly through a joystick direction.

It turns JoyDeck into a physical launcher, drastically reducing the need for mouse and keyboard when opening frequent programs and tools.

When to use it

Open App is ideal when you want to:

- quickly launch desktop applications;
- open professional or creative software;
- execute local scripts (.bat, .cmd);
- call applications via system or third-party URIs.

Parameters (Value)

The Value field contains the path or identifier of the application to launch.

Supported formats:

- Local path to an executable file (e.g., C:\Program Files\App\app.exe)
- System scripts (.bat, .cmd)
- System or application URIs (e.g., spotify:, ms-settings:, com.epicgames.launcher://)

Joystick behavior

- direction selection → visual feedback
- release / confirm → application launch
- JoyDeck does not check whether the app is already running

Visual feedback

- Illuminated LED segment → selected direction
- All LEDs on → command sent
- Red color → launch error

Practical examples

- Notepad++: C:\Program Files\Notepad++\notepad++.exe
- Script: C:\Scripts\start_work.bat
- URI: spotify:

Common mistakes

- Incorrect or incomplete path
- Missing file extension
- Using Open App for web links (use Open Link instead)

Notes and tips

- Prefer URIs when available
- Use absolute paths
- Assign recognizable icons

8.3 Open Link

Description

The Open Link Action allows opening web links using the system's default browser. It is intended exclusively for http and https URLs.

When to use it

Open Link is suitable for:

- websites and cloud services;
- dashboards and web panels;
- online documentation;
- support pages.

Parameters (Value)

The Value field must contain a valid URL starting with:

- http://
- https://

Example: <https://github.com>

Joystick behavior

When the command is confirmed:

- the link is opened in the default browser;
- JoyDeck does not check site availability;
- the action is instantaneous and non-persistent.

Visual feedback

- Illuminated LED segment → selected direction
- All LEDs on → link opened
- Red color → invalid URL

Practical examples

- <https://github.com>
- <https://docs.example.com>

Common mistakes

- Entering URLs without http/https prefix
- Using application URLs (in that case use Open App)

Notes and tips

- Use Open Link only for web content
- Always verify the URL prefix

8.4 System Control

Description

The System Control Action allows executing predefined system commands (e.g., shutdown, restart, lock session).

When to use it

It is suitable for:

- shutting down or restarting the PC;
 - locking the session;
 - performing quick system-level actions.
-

Parameters (Value)

The Value field contains the command selected from the list (it is not free text).

Typical examples:

- Shutdown
 - Restart
 - Lock
 - Logout
-

Joystick behavior

- direction selection → command preparation;
 - confirm / release → immediate execution of the command.
-

Visual feedback

- All LEDs on → command accepted
 - Red color → command not allowed or error
-

Practical examples

- Lock workstation
 - Restart
-

Common mistakes

- Assigning critical commands without a confirmation mode
- Accidental activations

8.5 Simulate Key

Description

The Simulate Key Action allows simulating keyboard input: single keys, combinations (shortcuts), sequences, text typing, and special modes such as Hold and Spam.

It is the most flexible Action in JoyDeck and covers most "power-user" automations without writing code.

When to use it

It is suitable when you want to:

- trigger shortcuts (e.g., CTRL+S, ALT+TAB);
- automate repetitive actions (sequences with waits);
- insert predefined text;
- manage "hold-to-press" keys (push-to-talk, sprint, etc.);
- repeat a key at intervals (spam) while holding the direction.

Choosing between Preset and Custom

In the Commands tab, Simulate Key can be used in two modes:

- Preset: select a predefined combination from the combo box (fast and syntax-safe).
- Custom: manually write a sequence in the textbox to create custom macros.

In general:

- use Preset for common shortcuts;
- use Custom for multi-step macros, text, and advanced automation.

Parameters (Value)

The Value field depends on the selected mode:

- Preset → Value is the name of the predefined choice (e.g., "Copy", "Paste", "Save"...)
- Custom → Value is a string describing the macro

Custom syntax (basics)

- Combinations: CTRL+S, CTRL+SHIFT+ESC
- Sequences: separate steps with commas (,)
Example: CTRL+C, CTRL+V
- Waits (if supported): WAIT 300 (milliseconds)
Example: CTRL+S, WAIT 250, ENTER
- Text (if supported): TEXT:...
Example: TEXT:DGVeLab

Note: if a token (key or command) is not recognized, the macro may fail.

Hold

The Hold mode is used when you want a key (or combination) to remain "active" persistently.

Unlike a normal press, Hold works like a toggle:

- first activation of the direction → the key is pressed and held;
- second activation of the same direction → the key is released and Hold ends.

This is useful when you don't want to physically keep the direction active but need a key held for a period.

Use cases:

- Gaming: keep SHIFT (sprint) or CTRL (crouch) active until disabled.
- Applications: keep a modifier (ALT, CTRL) pressed during a specific workflow.

- Push-to-talk: use Hold only if you prefer a “mute/unmute” toggle instead of “press to talk”.

Note: if the application loses focus or intercepts input, you may need to deactivate and reactivate Hold to realign the state.

Spam

The Spam mode automatically repeats a key or combination at regular intervals while you hold the direction.

It consists of:

- key/combination to repeat (preset or custom)
- repetition interval (in milliseconds)

Recommended intervals:

- 80–120 ms: fast repetition (games / scrolling)
- 150–250 ms: “human-like” and more compatible
- 300+ ms: slow repetition (delicate actions)

If the interval is too low, some applications may:

- miss events;
- ignore presses;
- behave unpredictably.

Joystick behavior

- On release: executes the macro when the direction is released.
- On confirm: selects the direction and confirms with the button.
- Hold: performs “press & hold” while the direction remains active.
- Spam: repeats while the direction remains active, respecting the interval.

In all cases, the simulation acts on the foreground window.

Visual feedback

- Illuminated LED segment → selected direction
- All LEDs on → macro sent / mode active
- Red color → invalid syntax or non-executable command

Practical examples

Preset examples

- Save → equivalent to CTRL+S
- Copy → equivalent to CTRL+C
- Paste → equivalent to CTRL+V

Simple Custom examples

- Save: CTRL+S
- New browser tab: CTRL+T
- Task Manager: CTRL+SHIFT+ESC

Advanced Custom examples

1. Windows search and program launch (typical example)
WIN, WAIT 200, TEXT:cmd, WAIT 100, ENTER
2. Save As and confirm (useful in editors)
CTRL+SHIFT+S, WAIT 300, ENTER

3. Macro "copy → go to end of line → paste on new line" (text editor)
CTRL+C, END, ENTER, CTRL+V
4. Spam for game / gathering (hold direction to repeat)
Key: E
Interval: 120 ms

Common mistakes

- Typing an unsupported key name (e.g., typo: CTLR instead of CTRL).
- Macro too fast: missing waits between windows or menus.
- Using Simulate Key when the correct window is not in the foreground.
- Spam interval too low.

Notes and tips

- Start with Preset and switch to Custom only when needed.
- For complex macros, insert small waits (WAIT) between steps.
- Always use Test before assigning critical macros.
- For Hold/Spam, choose comfortable directions and avoid overlap with important commands.

8.6 Mouse Move

The Mouse Move mode allows using JoyDeck as a mouse controller, turning the joystick into a Windows cursor pointing device, with support for click, double-click, and quick exit from the mode.

When a command is set to Mouse Move, JoyDeck enters a dedicated mode in which the behavior of the stick and button temporarily changes.

Activating Mouse Move mode

To activate Mouse Move:

1. Select a direction in the Commands panel
2. Set the Action to Mouse Move
3. Save the configuration
4. Execute the command through JoyDeck

Upon activation:

- JoyDeck enters Mouse mode
- Standard commands are suspended
- Stick movement directly controls the cursor

A confirmation message appears on screen:

Mouse mode active

Cursor movement

During Mouse Move mode:

- The analog stick controls cursor movement
- Speed is proportional to distance from center
- The dead zone prevents involuntary movement
- Response curves and acceleration are supported

Movement takes into account:

- Mouse sensitivity
- Mouse curve
- Mouse acceleration
- Axis inversion (internal to firmware/software)

The more the stick is tilted, the faster the cursor moves.

Clicks and joystick button

The joystick button performs the following functions:

Button action	Behavior
Single click	Left click
Double click	Right click
Long press	Exit Mouse Move mode

Clicks are simulated at the operating system level, as if coming from a physical mouse.

Exiting Mouse Move mode

To exit Mouse Move mode:

- Hold the joystick button

Upon exit:

- Mouse control is disabled
- JoyDeck returns to standard mode
- Any persistent keys or states are safely released

Displayed message:

Exited mouse mode

Mouse settings (Settings tab)

In the Settings tab, Mouse section, advanced parameters are available:

Mouse Sensitivity

- Adjusts overall cursor speed
- Higher values → faster movement

Mouse Curve

- Modifies non-linear response
- Values > 1 make movement more progressive and precise

Mouse Acceleration

- Adds dynamic acceleration
- Useful for covering large distances with small movements

These settings are applied in real time.

Important notes

- Mouse Move is an exclusive mode: while active, JoyDeck does not execute other commands
- Upon exiting the mode, JoyDeck performs a safety reset of:
 - simulated keys
 - active Spam
 - persistent Hold states
- Exit always occurs in a controlled manner to avoid stuck states or unwanted input

8.7 Orbital

Orbital is a persistent mouse-control mode designed to facilitate operations that require continuous button pressing, such as dragging, rotating, extended selections, or manipulations in graphic software, CAD, and 3D modeling tools.

Unlike Mouse Move, Orbital does not simply move the cursor: it automatically manages mouse-button press and release based on joystick movement.

Activating Orbital mode

To activate Orbital:

1. Select a direction in the Commands panel
2. Set the Action to Orbital
3. Configure the dedicated parameters
4. Save the configuration
5. Execute the command through JoyDeck

When the mode is activated:

- JoyDeck enters a dedicated persistent mode
- the selected mouse button is managed automatically
- standard commands are temporarily suspended

How it works

Orbital behavior is based on the stick's position relative to the central dead zone.

- When the stick leaves the dead zone, JoyDeck automatically presses the configured mouse button
- When the stick returns to the central zone, a release timer starts
- If the stick remains centered for the configured time, the button is released automatically
- If the stick is moved again before the timer expires, the button remains pressed

This allows smoother work in all situations requiring continuous dragging or a maintained press, without physically holding the joystick button.

Available parameters

Orbital mode provides two dedicated parameters.

Orbital button

Defines which mouse button is used during the mode.

Available options:

- Left
- Right
- Center

Release delay (ms)

Defines the waiting time before automatically releasing the button when the stick returns to the dead zone.

Recommended initial value:

- 1000 ms

Lower values make the release faster.

Higher values increase stability during small movements or short pauses.

Joystick behavior

During Orbital:

- stick movement continues to control the cursor
- leaving the dead zone triggers automatic pressing of the selected button
- returning to center does not immediately release the button, but starts the release timer
- leaving the dead zone again immediately restores the press

This logic makes Orbital particularly useful for continuous movements with a maintained press.

Exiting Orbital mode

To exit Orbital:

- hold the joystick button

Upon exit:

- the mode is deactivated
- any active mouse presses are safely released
- JoyDeck returns to standard operation

Visual feedback

During Orbital:

- a portion of the LEDs identifies the selected direction
- LEDs may remain active to indicate that the mode is running
- the color associated with the direction continues to represent the configured command

The purpose of the feedback is to make it immediately clear that JoyDeck is in a persistent mode different from normal behavior.

Practical examples

Orbital is particularly useful in scenarios such as:

- rotating the view in CAD software
- continuous object dragging
- extended selections in graphic editors
- navigation in 3D environments
- repeated operations requiring a held click

Notes and tips

- Orbital is a persistent mode: it should not be considered a simple click
- it is advisable to assign it to an easily reachable direction
- the release delay value should be adjusted according to your workflow
- for precision tasks, test the mode with medium values before lowering the delay

8.8 Media Control

Activating Media Control mode

To activate Media Control:

1. Select a direction in the Commands panel
2. Set the Action to Media Control
3. (Optional) Enter a parameter in the textbox
4. Save the configuration
5. Execute the command through JoyDeck

On first use:

- Media Control is activated
- JoyDeck enters multimedia controller mode
- Stick movements no longer execute standard commands

Displayed message:

Media controller active

Optional parameter in the textbox (link/app)

The Media Control command can work without parameters, but the textbox can be used to specify what to open/activate before using media controls.

You may enter:

- Open link (http/https) → opens a webpage (e.g., YouTube / Spotify Web)
- Open app (local path or URI) → launches an app or invokes a protocol (e.g., VLC, Spotify, etc.)

Examples:

Web links

- <https://www.youtube.com>
- <https://open.spotify.com>

App / local path

- C:\Program Files\VideoLAN\VLC\vlc.exe

URI / protocol

- spotify:
- ms-settings:

Note: the parameter is used to “bring you” directly into the correct app or content, after which JoyDeck controls play/next/volume using media commands.

Available controls

During Media Control mode, the stick and button take on dedicated functions.

Vertical movements

- Up → Volume up
- Down → Volume down

Horizontal movements

- Right → Next track
- Left → Previous track

If the horizontal movement is held, the Seek function (forward/backward) is activated, if supported by the active app.

Exiting Media Control mode

To exit Media Control mode:

- use a long press on the joystick button

Upon exit:

- Media mode is deactivated
- Internal states are reset
- Any active key simulations are released

8.9 Spotify

Description

The Spotify Action allows JoyDeck to open Spotify content and automatically enter a dedicated media-control mode optimized for the Spotify application.

Unlike generic media control, this Action integrates launching the application and controlling playback within a single workflow.

This allows JoyDeck to function as a physical Spotify controller, enabling quick access to playlists, albums, or tracks directly from the joystick.

When to use it

The Spotify Action is recommended when you want to:

launch Spotify quickly

open a specific playlist, album, or track

control playback without using the keyboard

create a dedicated music-control direction on JoyDeck

It is especially useful in workflows where background music is frequently managed.

Parameters (Value)

The Value field contains the Spotify resource to open.

JoyDeck supports two formats.

Spotify URI

Native Spotify URIs allow opening specific content directly in the Spotify application.

Examples:

- spotify:playlist:xxxxxxx
- spotify:track:xxxxxxx
- spotify:album:xxxxxxx
- spotify:artist:xxxxxxx

This is the preferred method when the Spotify desktop application is installed.

Web link

Standard Spotify web links are also supported.

Example:

- <https://open.spotify.com/playlist/xxxxxxx>

JoyDeck automatically forwards the link to the system, allowing Spotify to handle the request.

Automatic playback

After opening Spotify content, JoyDeck waits a few seconds for the application to initialize and then automatically sends a Play command.

This ensures that:

- the playlist starts immediately
- the user does not need to interact with the application manually

This delay prevents playback commands from being sent before Spotify is fully ready.

Spotify Controller Mode

After launching Spotify through this Action, JoyDeck enters Spotify Controller Mode. In this mode, the joystick can control playback directly.

Typical controls include:

- Vertical movements
- Up → Volume up
- Down → Volume down
- Horizontal movements
- Right → Next track
- Left → Previous track

The control logic is optimized specifically for Spotify behavior.

Exiting Spotify Mode

To exit Spotify Controller Mode:

1. hold the joystick button

Upon exit:

- playback remains active
 - controller mode is disabled
 - JoyDeck returns to normal operation
-

Visual feedback

Illuminated LED segment → selected direction

All LEDs on → Spotify controller active

Red color → error opening Spotify

A status message may appear on screen indicating that Spotify control mode has been activated.

Practical examples

Open a personal playlist:

- `spotify:playlist:37i9dQZF1DXcBWIGoYBM5M`

Open a specific track:

- `spotify:track:6rqhFgbbKwnb9MLmUQDhG6`

Open Spotify home:

- `spotify:`
-

Common mistakes

Using Open Link instead of Spotify for Spotify URIs.

Entering incomplete Spotify links.

Trying to control Spotify when playback is active on a remote device.

Notes and tips

Spotify commands work best when audio playback is set to This Computer.

If Spotify is streaming to another device (phone, speaker, smart TV), volume control may not behave as expected.

Using Spotify URIs instead of web links generally produces more reliable results.

8.10 HTTP Request

The HTTP Request command allows JoyDeck to send HTTP GET or POST requests to external services, APIs, webhooks, or automation systems.

It is designed for:

- integration with web services
- remote automation
- IoT systems
- triggering external events

The request is executed in the background without blocking the interface.

Command configuration

To configure an HTTP request:

1. Select a direction in the Commands panel
2. Set the Action to HTTP Request
3. Enter the parameter in the textbox
4. Save the configuration
5. Execute the command through JoyDeck

If the syntax is valid:

- the request is sent immediately
- JoyDeck only displays errors if they occur

Supported syntax

The textbox accepts a compact string in the following general format:

METHOD URL

or, for more advanced requests:

METHOD URL | BODY

Where:

- METHOD → GET or POST
- URL → full address (http / https)
- BODY → optional content (POST only)

Examples – HTTP GET

Simple GET

GET https://example.com/api/status

Sends a standard GET request.

GET with parameters

GET https://example.com/api/device?state=on&id=3

Useful for REST APIs or automation systems.

GET as trigger (webhook)

GET https://maker.ifttt.com/trigger/light_on/with/key/XXXXXXXX

JoyDeck acts as a remote physical button.

Examples – HTTP POST

POST without body

POST <https://example.com/api/start>

Used when the endpoint requires no data.

POST with text body

POST <https://example.com/api/message> | Hello from JoyDeck

Sends a simple string as content.

POST with JSON

POST <https://example.com/api/device> |

```
{  
  "device": "lamp",  
  "state": "on",  
  "brightness": 80  
}
```

JoyDeck sends the body exactly as written, without modification.

Integration with Home Automation / IoT

ESP / microcontrollers

GET <http://192.168.1.50/relay/on>

Home Assistant

POST <http://homeassistant.local:8123/api/services/light/toggle> |

```
{  
  "entity_id": "light.living_room"  
}
```

IFTTT

GET [https://maker.ifttt.com/trigger/door open/with/key/XXXXXXXXXX](https://maker.ifttt.com/trigger/door+open/with/key/XXXXXXXXXX)

System behavior

- The request is asynchronous
- JoyDeck does not wait for a visible response
- Network errors are logged
- The command is considered executed even without a response

In case of error:

- a message is displayed
 - the direction LED may blink red (if supported)
-

Important notes

- JoyDeck does not validate the body content
- The user is responsible for URL correctness
- Requests do not support advanced authentication (tokens, custom headers)
- For complex APIs, using an intermediate bridge (e.g., webhook / script) is recommended

9. Command Trigger Modes

JoyDeck allows commands assigned to joystick directions to be activated through different trigger modes. These modes determine when and how the associated command is executed.

The goal is to let the user adapt the device's behavior to their workflow, avoiding accidental activations and enabling both immediate commands and more persistent controls.

9.1 Immediate Trigger

This is the standard operating mode.

The command is executed immediately when the stick enters the assigned direction.

Main characteristics:

- instant activation
- ideal for quick shortcuts
- behavior similar to a physical button

Usage examples:

- opening an application
- executing a keyboard shortcut
- sending an HTTP request
- opening a Spotify playlist

When the stick returns to the neutral position, no additional action is executed.

9.2 Controller Mode

Some actions activate a persistent control mode in which JoyDeck becomes a dedicated controller for a specific function.

In this mode, the stick does not execute a single command but continuously controls an active function.

Examples:

Mouse Mode

JoyDeck becomes a mouse controller.

The stick controls:

- cursor movement
- speed proportional to distance from center

This mode is useful for:

- fast navigation
- presentations
- controlling applications without a keyboard

Media Controller Mode

JoyDeck controls media playback.

The stick can manage:

- Play / Pause
- next track
- previous track
- volume

- forward/backward seek

Behavior may vary depending on the active application (browser, Spotify, video player, etc.).

Spotify Controller Mode

When Spotify is opened through JoyDeck, the device can automatically enter the dedicated controller mode. In this mode, the stick directly controls Spotify.

9.3 Orbital Mode

Orbital mode is a persistent mode designed to simulate the behavior of a mouse button held down during movement.

This mode is particularly useful for operations such as:

- rotating 3D objects
- navigating CAD software
- manipulating the camera in 3D environments

How it works:

- The stick is moved into a programmed direction.
- When the stick exits the central dead zone:
JoyDeck automatically presses a mouse button.
- When the stick returns to the neutral zone:
a release timer starts.
- If the stick remains still in the neutral zone for the configured time:
the mouse button is released automatically.

Configurable parameters:

- mouse button used
 - Left
 - Right
 - Center
- release time (ms)

This mode allows continuous control without physically holding a button.

9.4 Safety and State Management

JoyDeck includes safety mechanisms to prevent:

- simulated keys from remaining pressed
- controller modes from staying active unintentionally

When an action that changes context is executed (such as opening an application), the system automatically releases any simulated keys still active.

This behavior ensures stability and prevents situations where a key combination remains stuck in the operating system.

10. Saving, Importing / Exporting Configurations

JoyDeck Hub allows saving, exporting, and importing custom configurations.

A configuration contains:

- actions assigned to directions
- action parameters
- device settings
- operating modes

This system makes it possible to:

- create multiple profiles
- share configurations
- back up settings

10.1 Saving the configuration

To save the current configuration:

1. Open the File menu
2. Select Save

The program automatically stores the current settings in the local configuration file.

Saving is useful when:

- command assignments are modified
- an action's parameters are updated
- new operating modes are configured

10.2 Exporting the configuration

Exporting allows creating a configuration file to use as a backup or to share with other users.

Procedure:

1. Open the File menu
2. Select Export Config file
3. Choose the save location

A file containing all current settings will be created.

Typical uses:

- configuration backups
- transferring settings to another computer
- sharing presets

10.3 Importing a configuration

You can load a previously exported configuration.

Procedure:

1. Open the File menu
2. Select Import Config file
3. Select the configuration file

JoyDeck Hub will immediately load the settings contained in the file.

Current settings will be replaced by the new configuration.

10.4 Tips for configuration management

For efficient profile management, it is recommended to:

- create dedicated configurations for different use cases
 - productivity
 - CAD / 3D modeling
 - media control
 - presentations
- keep backup copies of important configurations
- use descriptive names for exported files

11. FAQ & Troubleshooting – Commands

A command is not executed

Possible causes:

- The Action is set to No Action
- The textbox is empty for a command that requires a parameter
- The command was not saved
- JoyDeck is not connected to the device

Solution:

- Check the command type and its value
- Press Save after every change
- Verify the connection status

The LED lights up but nothing happens

Possible causes:

- The command is correctly configured but:
 - the app is not installed
 - the link is unreachable
 - the Action requires an active window

Solution:

- Check the app path or URL
- Verify that the application is available on the system
- Check the JoyDeck Hub logs

Mouse Move does not work

Possible causes:

- The operating system is blocking mouse control
- The window does not have the required permissions

Solution:

- Run JoyDeck Hub as a standard user
- Verify that the system allows input control (Java Robot)
- Exit the mode with a long press

Media Control does not respond

Possible causes:

- The application does not support Windows media keys
- The correct window is not in the foreground

Solution:

- Bring the media application to the foreground
- Use the textbox parameter to open the app before controlling it

Spotify responds only partially

Possible causes:

- Spotify is playing on a remote device
- Audio output is not local

Solution:

- Switch playback to "This computer"
- Reactivate the Spotify command
-

Simulate Key remains "stuck"

Possible causes:

- Use of persistent keys (*)
- SPAM mode active

Solution:

- Use a direction with Simulate Key to deactivate
- Switch to another mode (Mouse / Media)
- Disconnect JoyDeck (automatic safety release)

HTTP Request has no effect

Possible causes:

- Incorrect or unreachable URL
- Unsupported method
- Server offline

Solution:

- Test the URL in a browser
- Use only GET or POST
- Verify that the service accepts external requests

A command works only once

Possible causes:

- On confirm mode is active
- Confirmation timeout expired
- Hold mode not completed

Solution:

- Press the button to confirm
- Hold the stick for the required time
- Verify the trigger mode

JoyDeck appears "unresponsive"

Possible causes:

- A special mode is active (Mouse / Media / Spotify)
- A direction is locked waiting for return to center

Solution:

- Return the stick to the center
- Use a long press to exit the active mode

12. Advanced FAQ – Power Users & Automation

Can I combine multiple actions into a single command?

Yes, indirectly.

JoyDeck executes one Action per direction, but you can chain behaviors using:

- Simulate Key (macro with WAIT)
- HTTP Request toward scripts or intermediate webhooks
- Media Control / Spotify with app launch via parameter

Practical example:

1. HTTP Request → triggers a webhook
2. Webhook → runs a script that launches apps, changes states, etc.

JoyDeck is intentionally kept simple; complex logic should be handled “downstream”.

Is it possible to send multiple HTTP Requests at once?

Not directly from the textbox.

Each HTTP command sends a single request.

For multiple requests:

- use an endpoint that performs several actions
- or a bridge (Node-RED, Python script, Home Assistant, etc.)

Recommended:

GET http://localhost/joydeck/scene/desk_on

→ the scene handles everything.

Does JoyDeck support HTTP authentication (tokens, headers)?

No, not natively.

Intentional limitations:

- no custom headers
- no bearer tokens
- no OAuth

Recommended solution:

- create a proxy endpoint that handles authentication
- JoyDeck calls only the “clean” endpoint

Can I use JoyDeck as a local IoT controller (LAN)?

Yes, and it's one of the most reliable use cases.

JoyDeck works very well with:

- local IPs
- ESP / Raspberry
- lightweight HTTP servers

Example:

GET <http://192.168.1.50/light/toggle>

Advantages:

- no cloud dependency
- minimal latency
- high reliability

Is it possible to distinguish short and long presses?

Yes, through Trigger Mode.

JoyDeck supports:

- On release
- On confirm
- Hold

This allows:

- different actions on the same direction
- prevention of accidental commands
- "layered" joystick usage

Can I use Simulate Key for games or professional software?

Yes, with some considerations.

It works well with:

- keyboard shortcuts
- CAD / editing software
- games not protected by anti-cheat systems

Limitations:

- some games block simulated input
- very fast macros may be filtered

Suggestions:

- use WAIT
- avoid extremely fast spam
- always test with the target software

What happens if I change mode while a key is being "held"?

JoyDeck always applies a safety release.

Any context change:

- Mouse mode
- Media Control
- Spotify
- Open App / Link

Triggers:

- release of all simulated keys
- stop of SPAM
- reset of internal states

No input ever remains stuck.

Is it possible to control JoyDeck from external software?

Not directly.

JoyDeck is designed as:

- a local endpoint
- not as a server

For advanced integration:

- use JoyDeck → HTTP → external software
- not the other way around

Can JoyDeck replace a full automation software?

No, and that is not its purpose.

JoyDeck is:

- an intelligent physical controller
- a reliable trigger
- a human–automation bridge

For complex logic:

- Home Assistant
- Node-RED
- custom scripts

JoyDeck remains the immediate control point.

13. Final Note

JoyDeck Hub always applies safety mechanisms:

- automatic key release
- forced exit from active modes
- prevention of stuck inputs

In case of abnormal behavior, disconnecting and reconnecting JoyDeck Hub safely restores the system state.